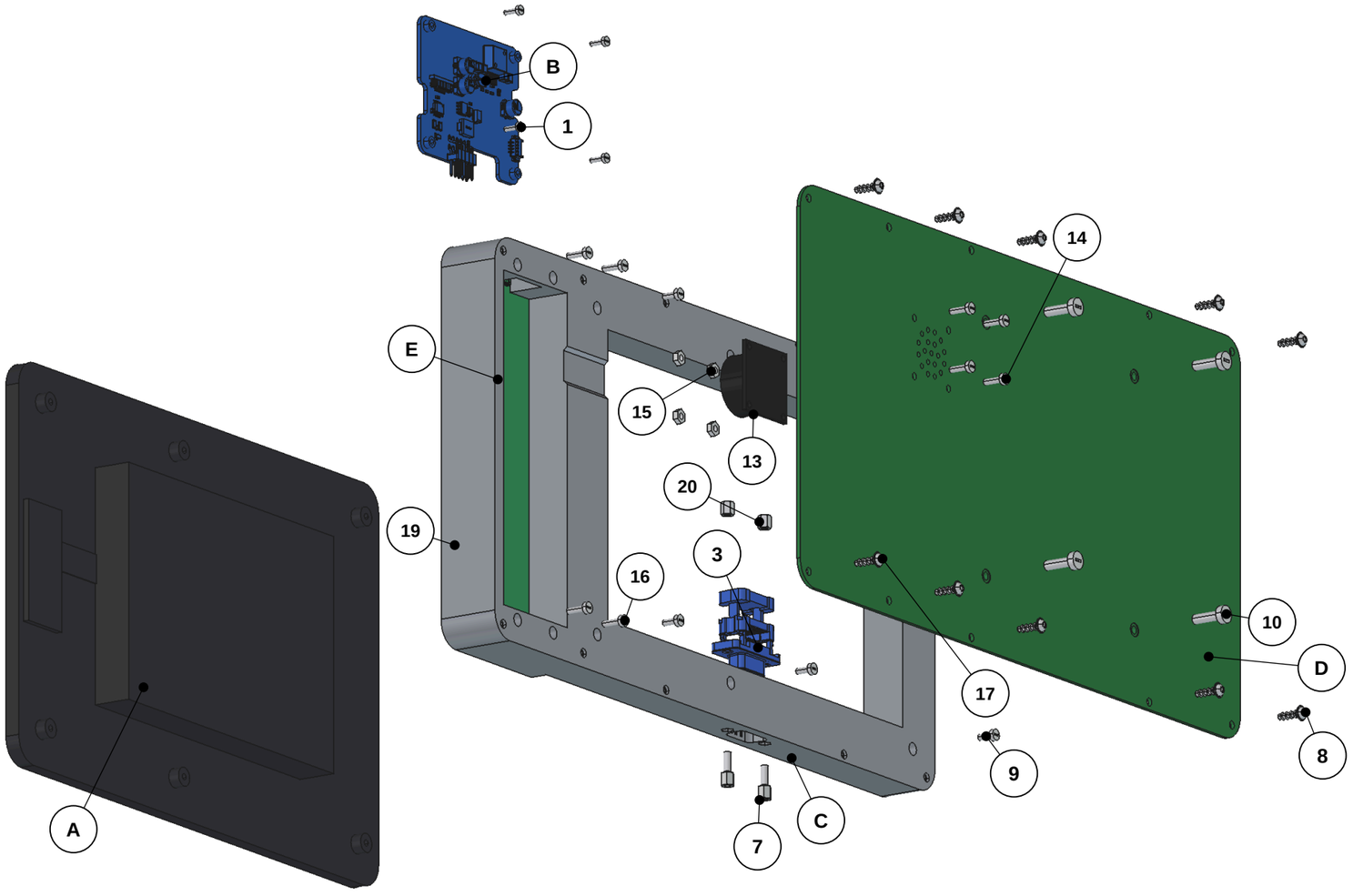
Exploded view, seen from behind



Parts list

Pos	Qty	Part / manufacturer, MPN
A	1	Display Guition JC8012P4A1C, 10.1", ESP32-P4, with display cover
B	1	CAN addon board pcbs/can_power/, under the display cover on S1-S4
C	1	Frame with keypad chamber, 3D printed, mechanics/frame_keypad.stl
D	1	Back plate PCB pcbs/backplate_keypad/, with VESA spacers H1-H4
E	1	Keypad PCB pcbs/keypad/ (keys F1-F12, spacers H1-H4)
1	4	Screw M2x6
2	1	FFC 14 pin, 0.5 mm pitch, 50 mm, type 2 (contacts on opposite sides) - Wurth Elektronik, 687714050002
3	1	D-Sub 9 female, IDC for 1.27 mm ribbon cable, without nuts - Wurth Elektronik, 618009221923
4	1	IDC socket 2x5, 2.54 mm, with strain relief (WR-BHD) - Wurth Elektronik, 61201023021
5	1	Coding plug for IDC socket
6	1	Ribbon cable 9 wire, 1.27 mm pitch, 28 AWG, 1 m piece - Wurth Elektronik, 63910915521CAB
7	2	Jackpost (hex standoff) UNC 4-40 inner/outer, hex 5.0, thread 8.2, SW 4.5 (WA-HEX) - Wurth Elektronik, 94513200
8	8	Thread forming screw for thermoplastics, WUEPLAST WN 1451, 3x10, pan head with flange (dk 6 mm), TX10 - Wurth, 4234 3 10
9	6	Screw M2.5x6 (original frame: M2.5x8)
10	4	Screw M4x12
11	2	WR-WTB crimp housing 1.25 mm, 4 pin (female) - Wurth Elektronik, 653004113322
12	2	WR-WTB pre-crimped wire 1.25 mm, contacts on both ends - Wurth Elektronik, 653153128130
13	1	Speaker 4 Ohm 3 W, 27.5 x 27.5 x 13 mm, cable 120 mm with 2-pin 1.25 mm plug - XHXDZ-2828
14	4	Screw M2.5x8
15	4	Nut M2.5
16	4	Screw M2.5x8
17	2	Thread forming screw for thermoplastics, WUEPLAST WN 1451, 3x10 - Wurth, 4234 3 10
18	1	FFC 14 pin, 0.5 mm pitch, about 250 mm, type TBD - Wurth Elektronik
19	1	PU foil printed + 3 mm plexiglass, 48 x 150 mm - Folika
20	2	Hex nut UNC 4-40, SW 4.5, 5 mm high (WA-HEX) - Wurth Elektronik, 94505009

Grey: not shown in the views. Pos. 1-20 as in assembly_bom.csv (notes there).

Order of assembly:

0 First: glue jackposts 7 into the hex pockets of C from outside (2K epoxy on the side walls only, female thread kept clean with a screw), let it cure (up to 24 h).
1 B onto the display board (cover off), screws 1; FFC 2 to FPC3, cable 11/12 to CN2.
2 Cover back on. C onto A, screws 9 (6x). E with 19 into the pocket of C, screws 16.
3 D-Sub 3 from inside over the jackposts 7, nuts 20 (SW 4.5, only about 0.1 Nm, secure with locking lacquer). FFC 18 from FPC4 to J1 of E.
4 Speaker 13 onto D (screws 14, nuts 15), plug into CN3.
5 D onto C, screws 8 and 17.
6 VESA bracket onto the spacers of D, screws 10.

JC8012P4A1C_I_W_Y_CAN
Assembly drawing

Variant 2: with keypad (12 keys F1-F12)

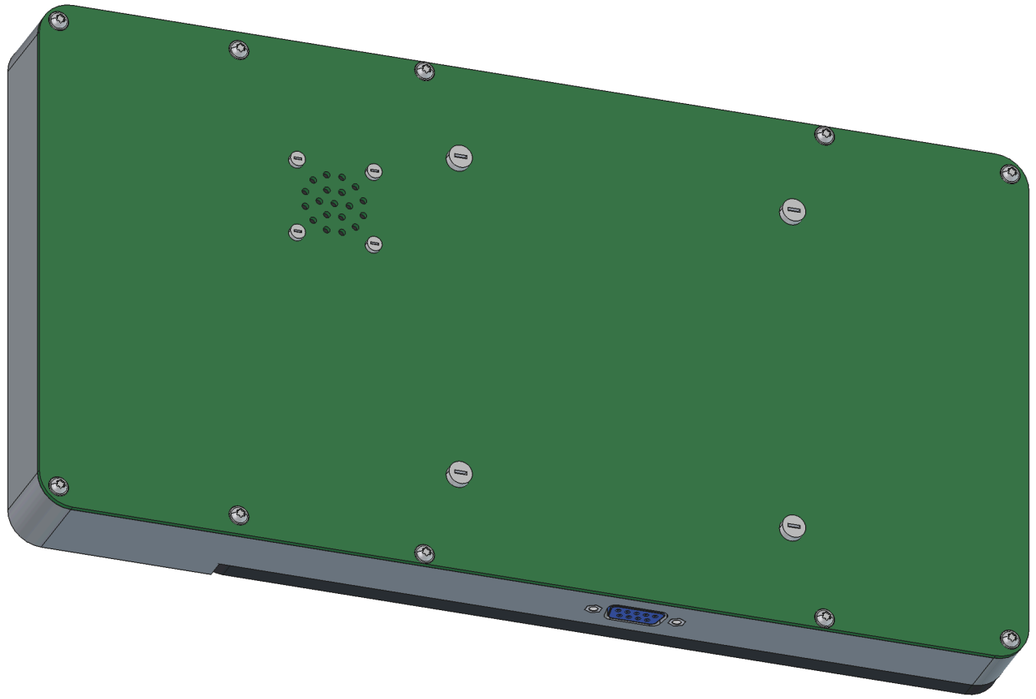
Meisterschulen am Ostbahnhof München
not to scale, dimensions in mm

Date 2026-11-13

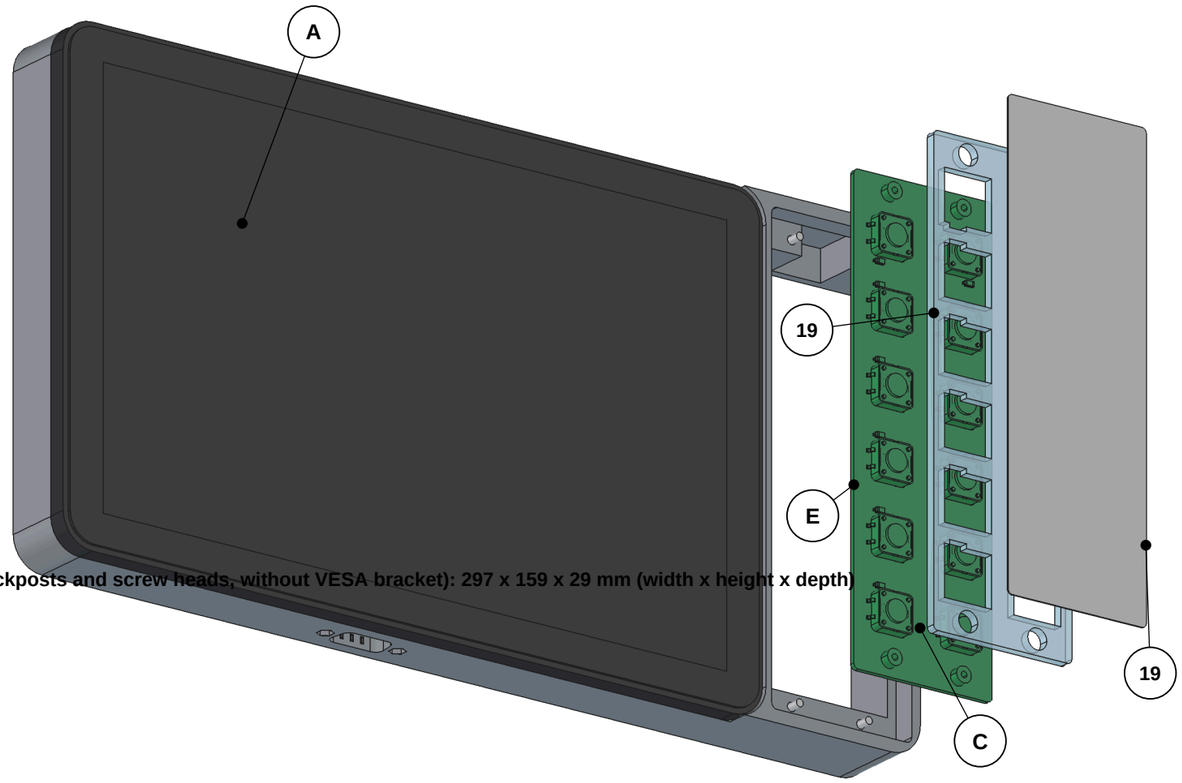
Assembled, front



Assembled, back



Keypad: E and 19 into the pocket of the frame from the front



Overall size (with jackposts and screw heads, without VESA bracket): 297 x 159 x 29 mm (width x height x depth)

JC8012P4A1C_I_W_Y_CAN

Assembly drawing

Variant 2: with keypad (12 keys F1-F12)

Meisterschulen am Ostbahnhof München
not to scale, dimensions in mm

Date 2026-11-13